SOFTWARE ENGINEERING LAB – 2

Agile Backlog Creation & Sprint Simulation in Jira

Dropshipping Inventory & Order Orchestrator

Submitted by

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Section:B

Jira Project: Dropshipping Inventory & Order Orchestrator

1. Project Overview

The objective of this lab was to create and simulate an Agile project backlog using Jira for a Dropshipping Inventory & Order Orchestrator. The system was divided into functional Epics and smaller User Stories so that the work could be planned, estimated, assigned to sprints, and tracked through a Scrum workflow.

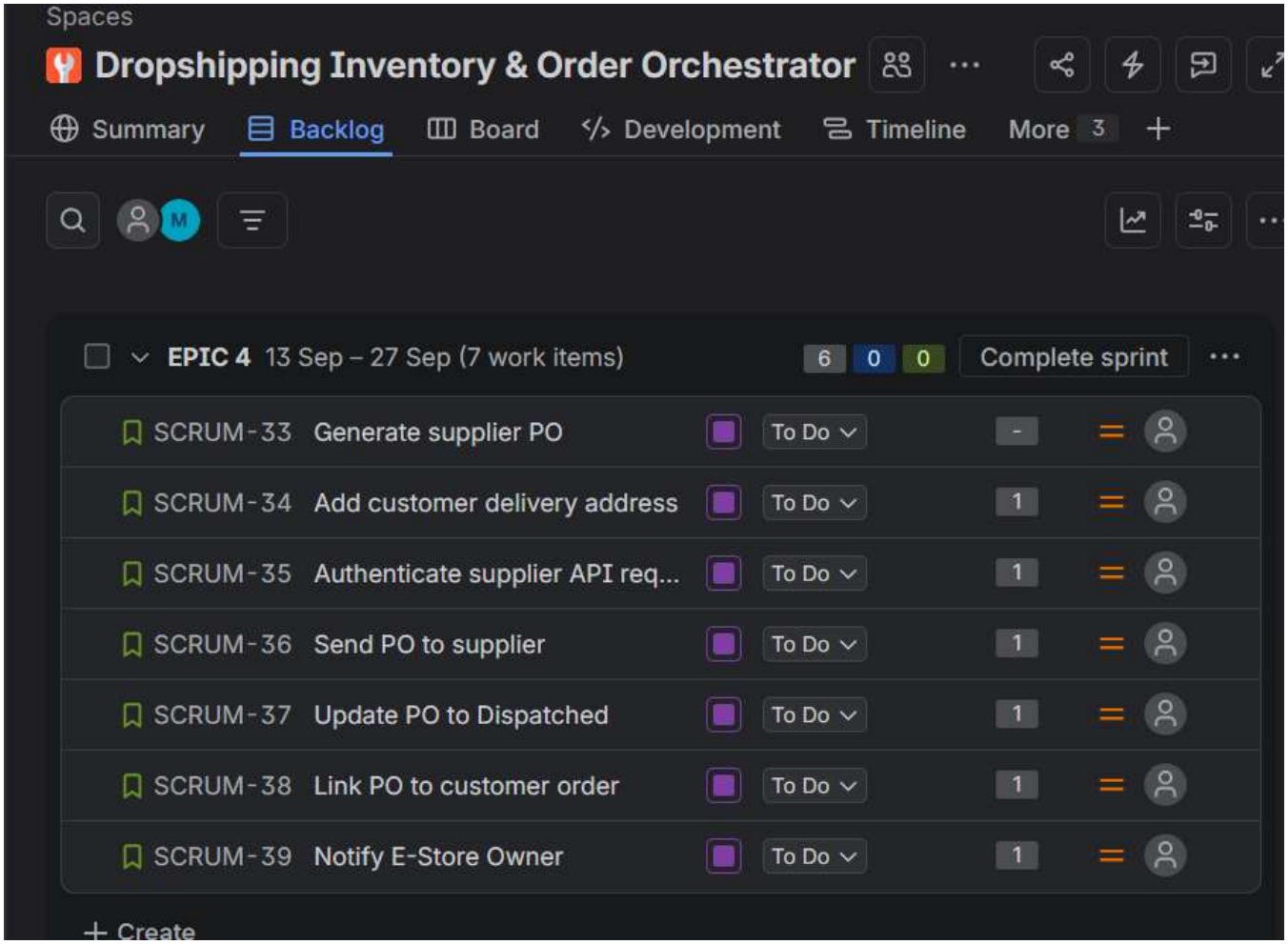
The implemented Jira workflow used the states To Do, In Progress, In Review, and Done. Separate sprints were created for the major Epics, and story-point estimates were assigned to the child User Stories. Jira's Active Sprint board and Burndown reports were then used to demonstrate progress and remaining sprint work.

Project Epic Structure

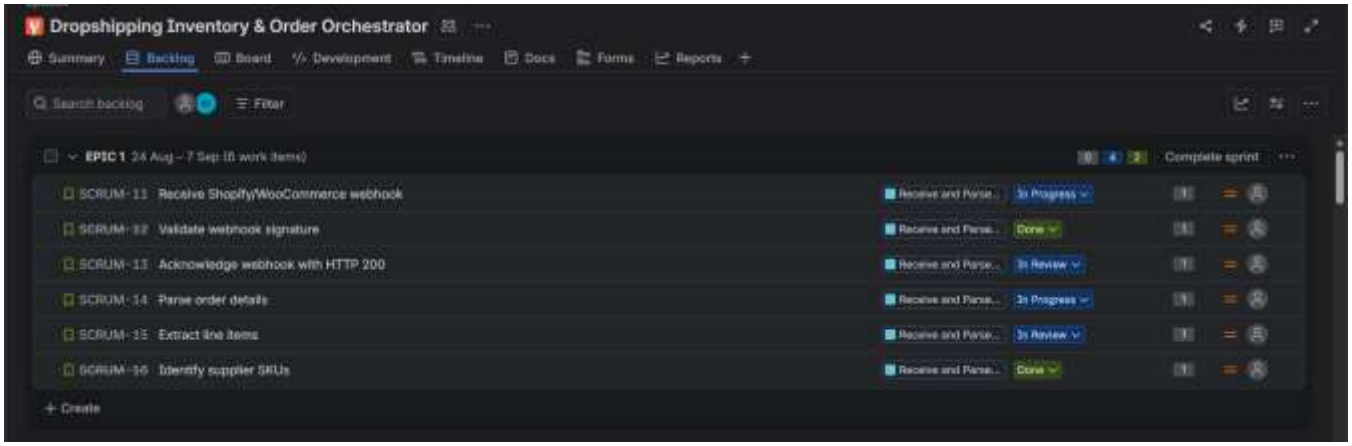
- Epic 1 – Receive and Parse Storefront Order Webhook
- Epic 2 – Verify Supplier Inventory Availability
- Epic 3 – Split Multi-Vendor Customer Order
- Epic 4 – Dispatch Supplier Purchase Orders
- Epic 5 – Track Supplier Shipment
- Epic 6 – View Order and Fulfillment Status

2. Jira Backlog with Epics and User Stories

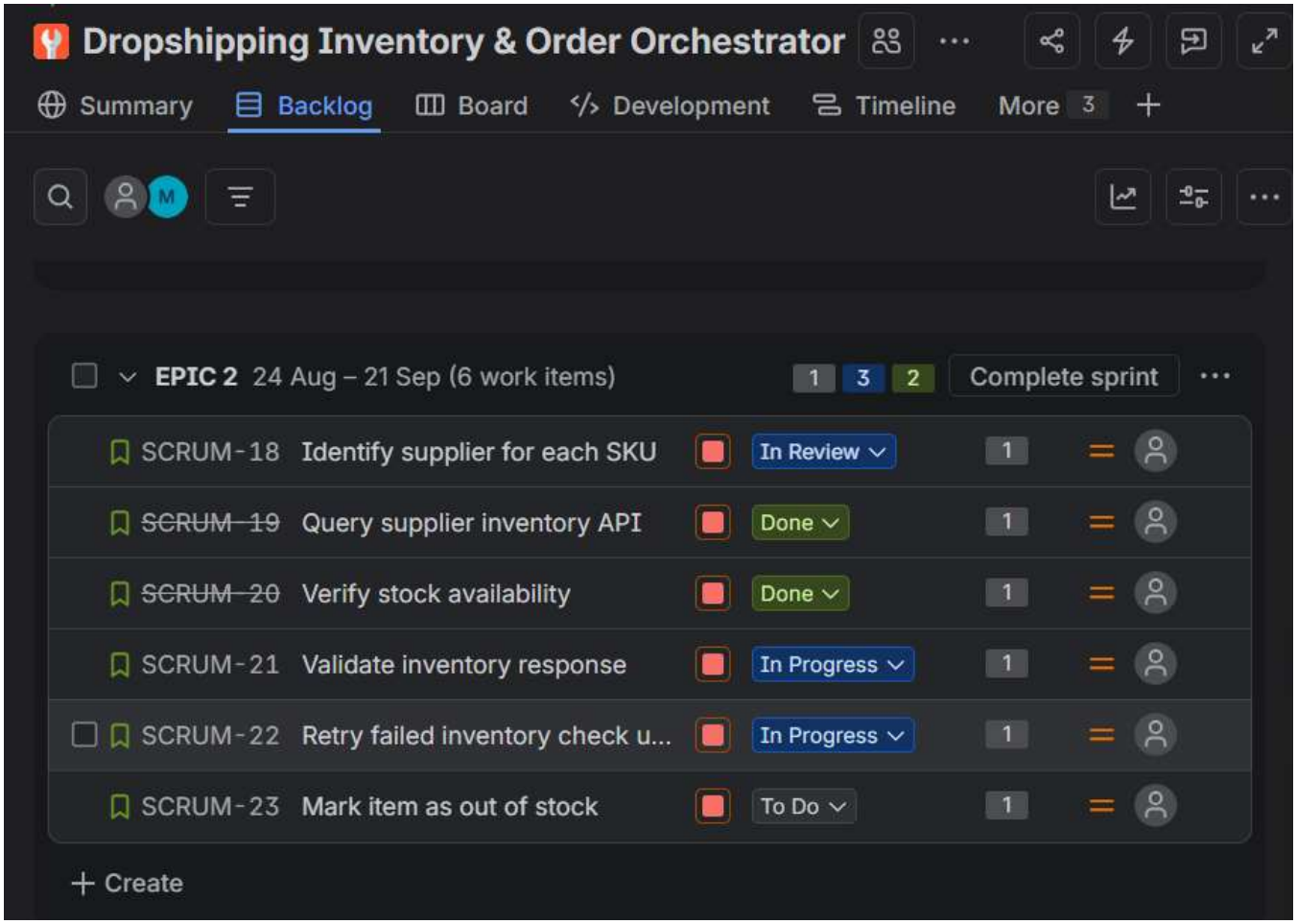
The backlog was organized into Epics with child User Stories. The screenshots below provide evidence of the Epic structure, sprint grouping, task status, and story-point estimates.



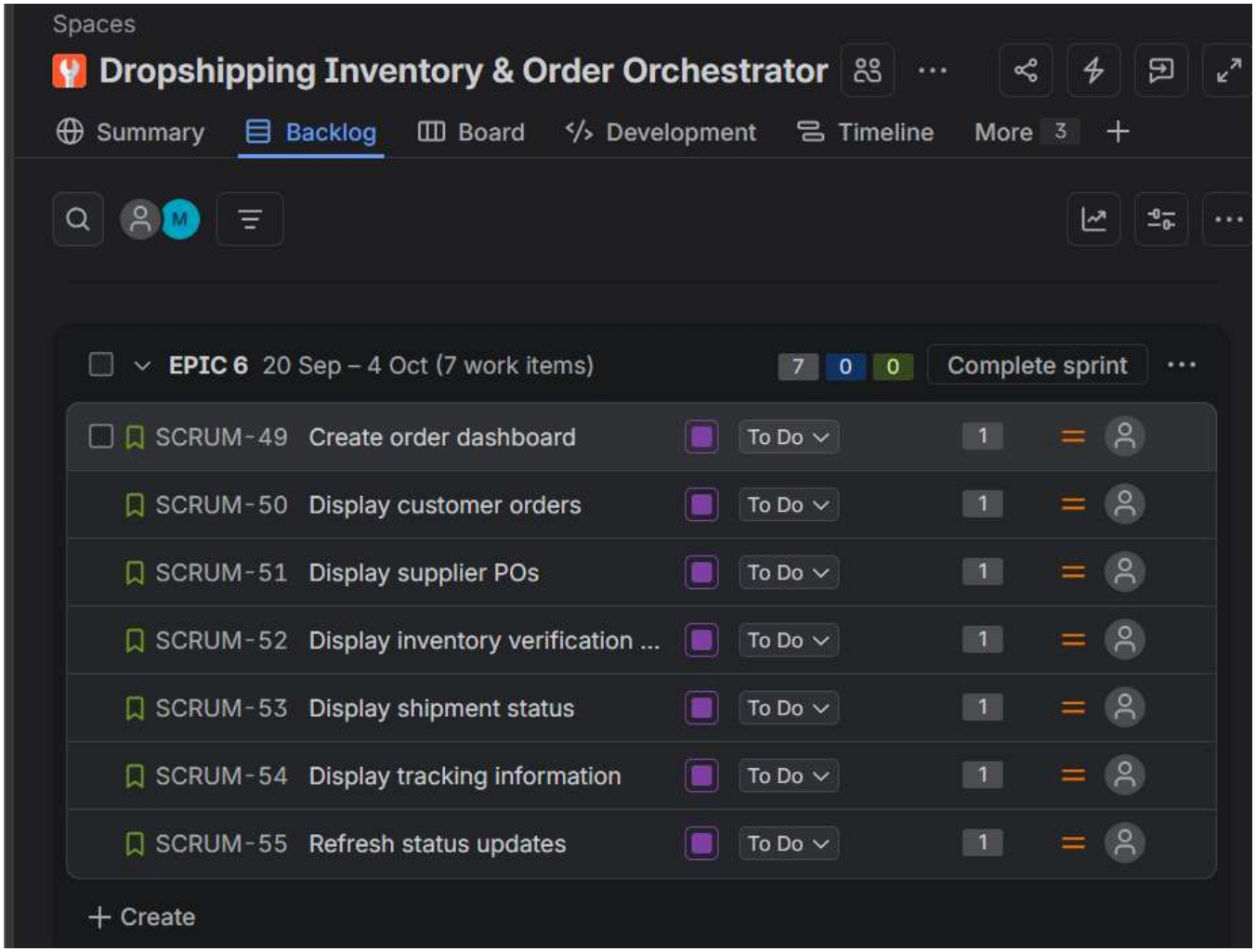
Backlog Evidence 1: Jira Epic/User Story organization



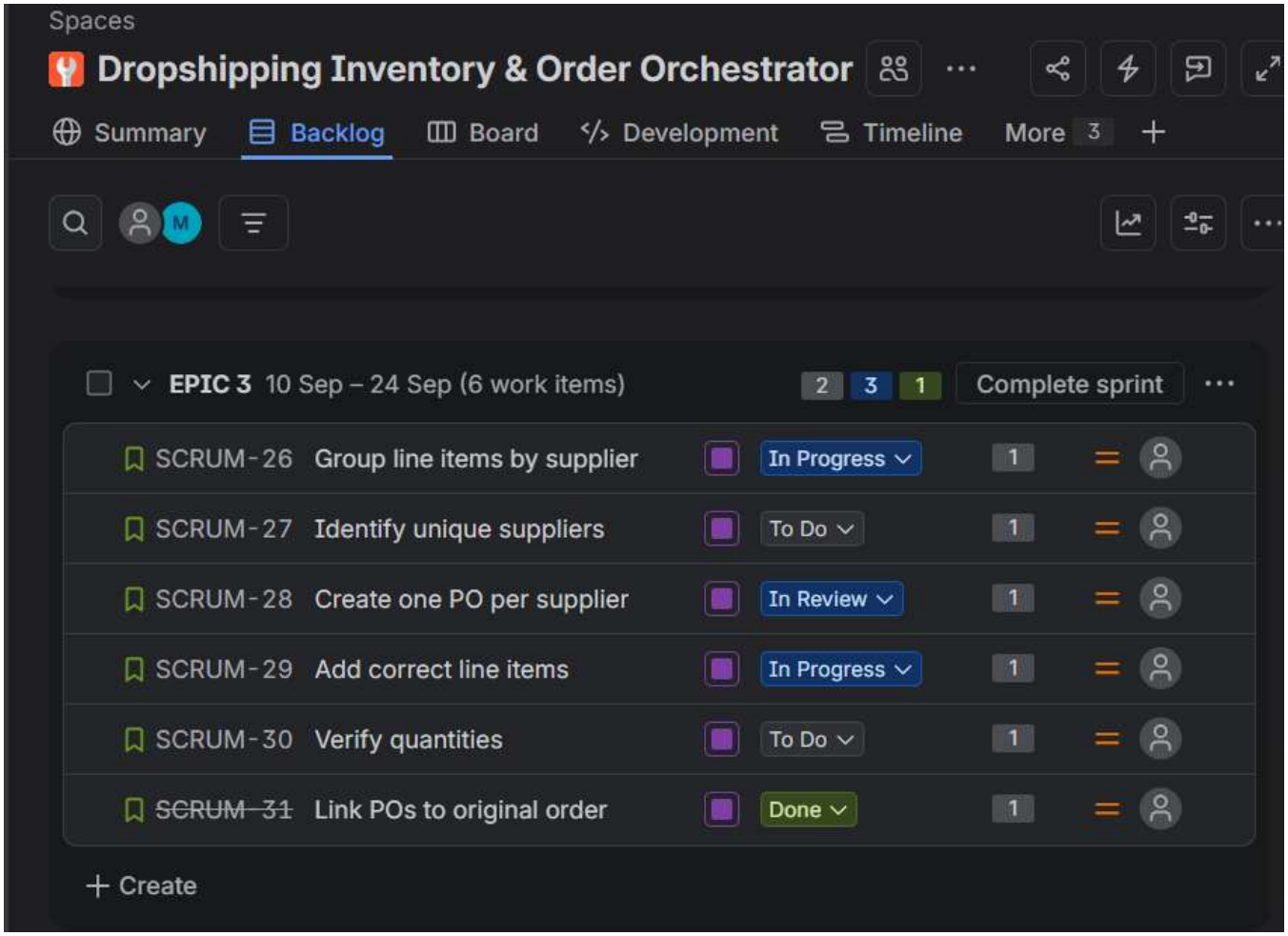
Backlog Evidence 2: Jira Epic/User Story organization



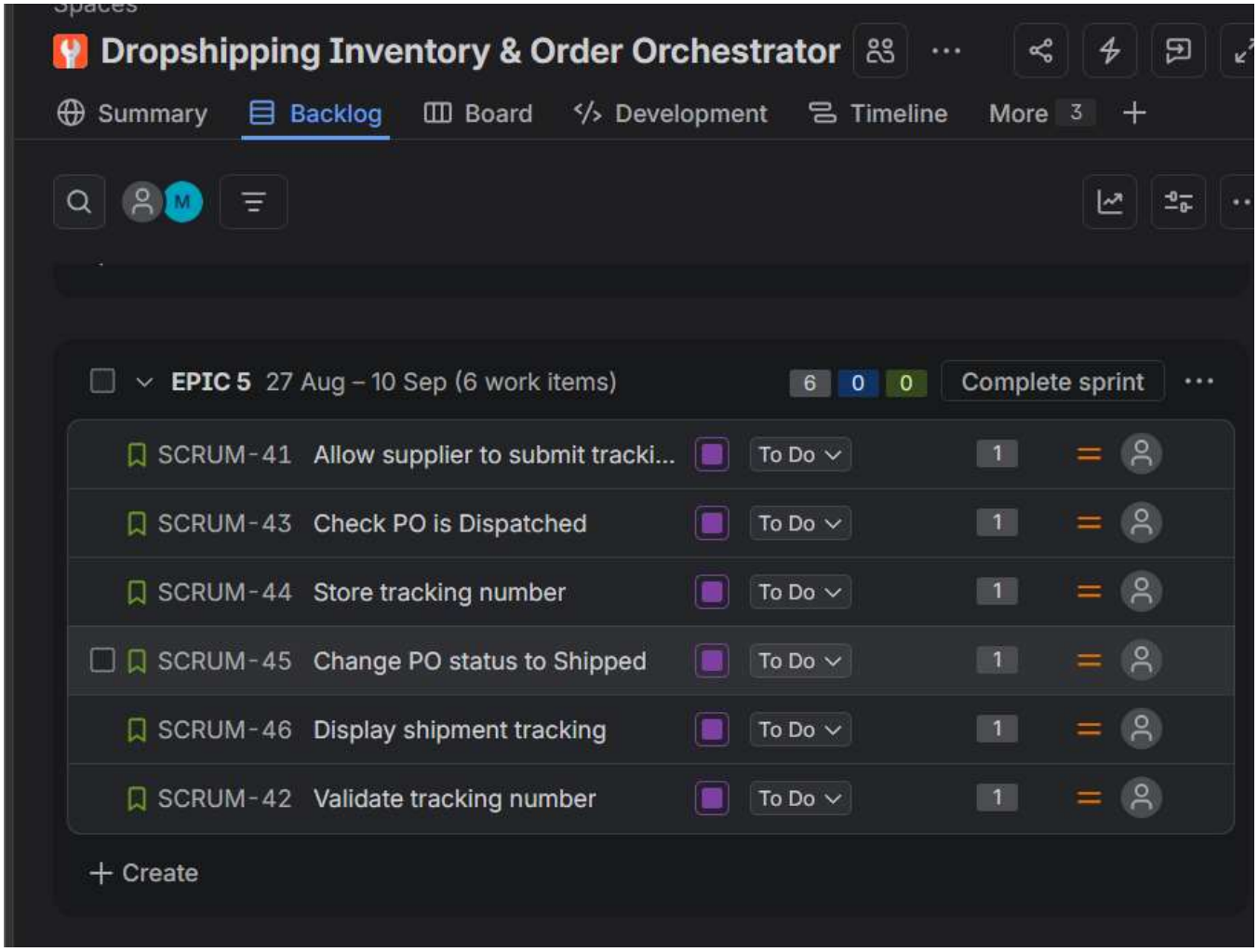
Backlog Evidence 3: Jira Epic/User Story organization



Backlog Evidence 4: Jira Epic/User Story organization



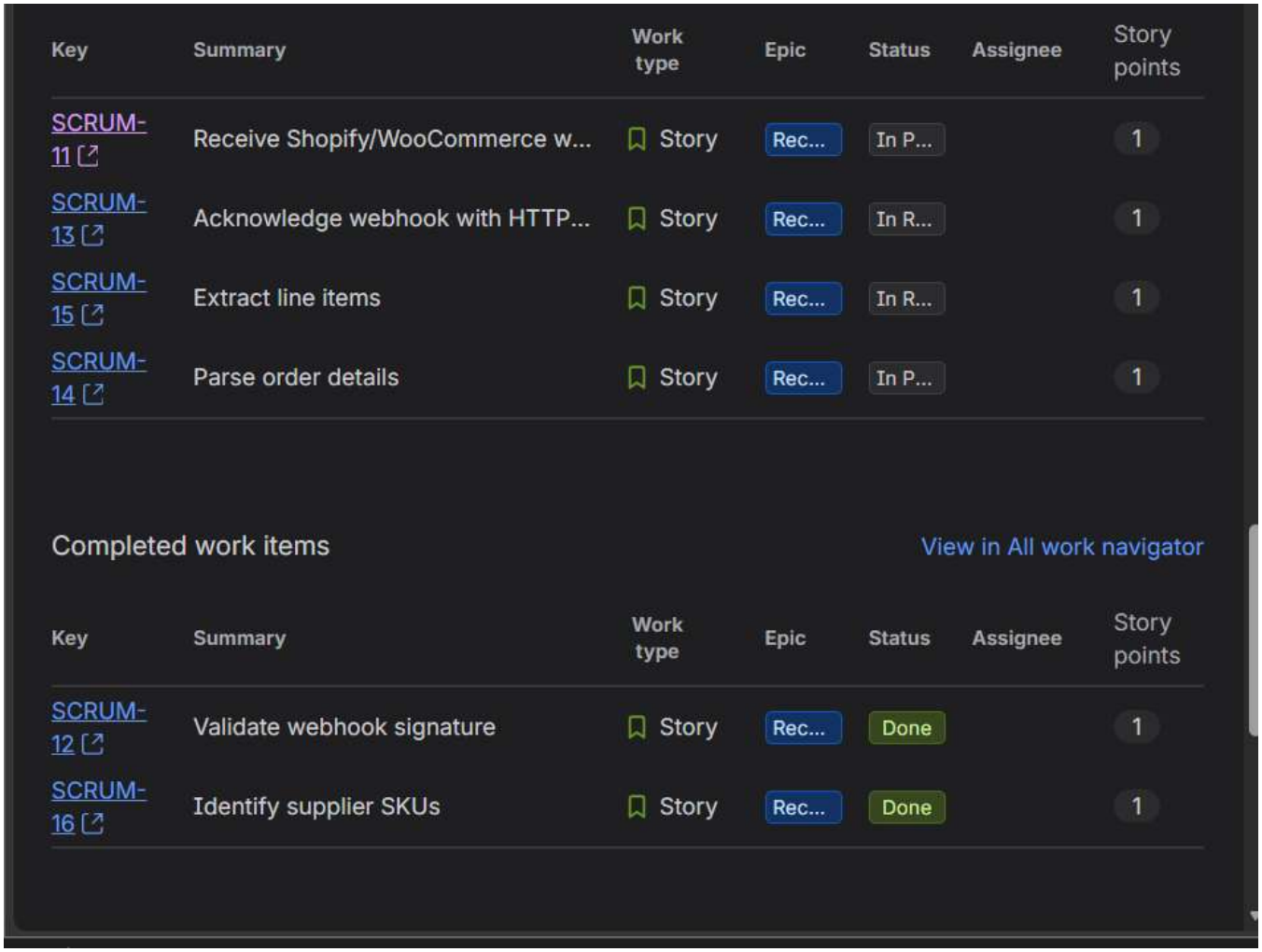
Backlog Evidence 5: Jira Epic/User Story organization



Backlog Evidence 6: Jira Epic/User Story organization

3. Story Point Assignments

Story points were used as relative estimates of effort. The User Stories were divided into small, focused tasks and, for the completed setup, the child Stories were assigned 1 story point each. The following screenshots show the estimates in Jira.

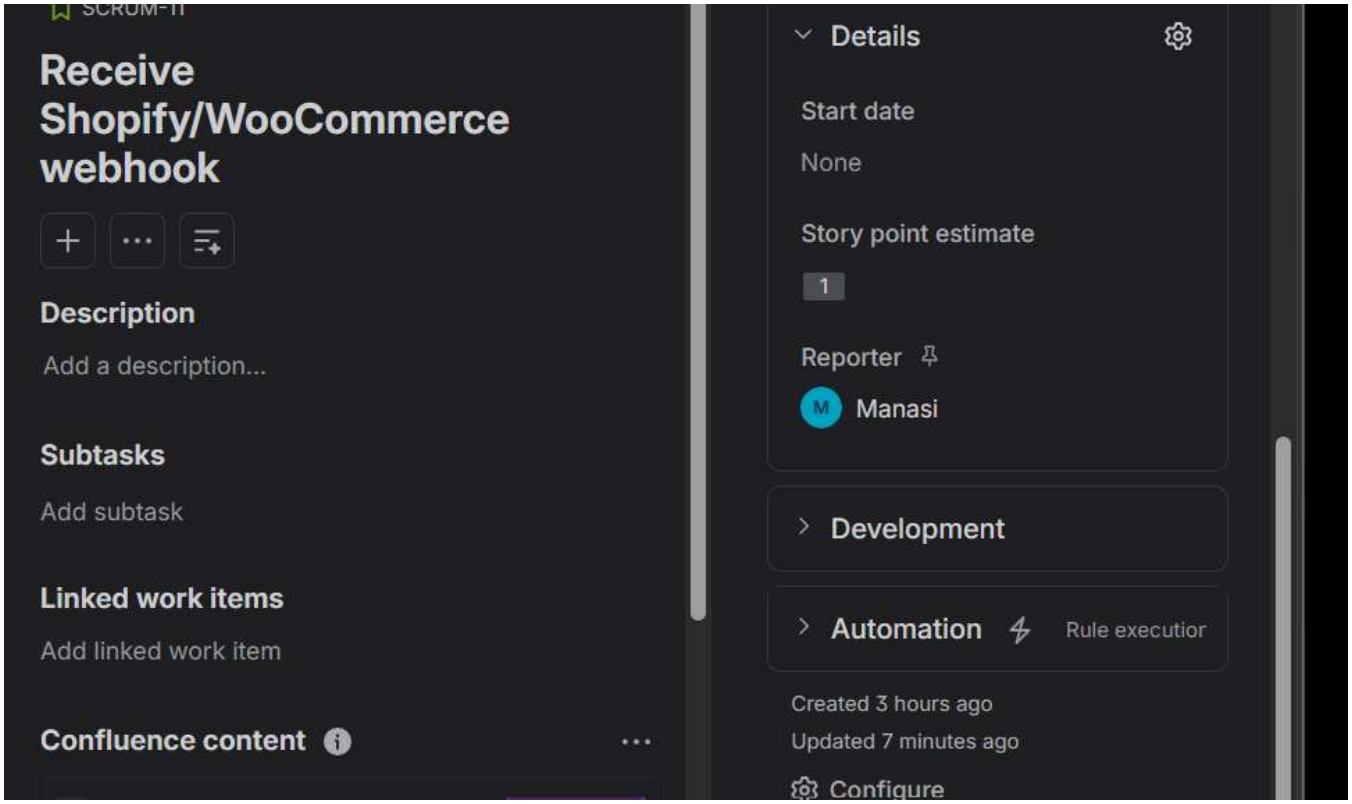


Key	Summary	Work type	Epic	Status	Assignee	Story points
SCRUM-11	Receive Shopify/WooCommerce w...	Story	Rec...	In P...		1
SCRUM-13	Acknowledge webhook with HTTP...	Story	Rec...	In R...		1
SCRUM-15	Extract line items	Story	Rec...	In R...		1
SCRUM-14	Parse order details	Story	Rec...	In P...		1

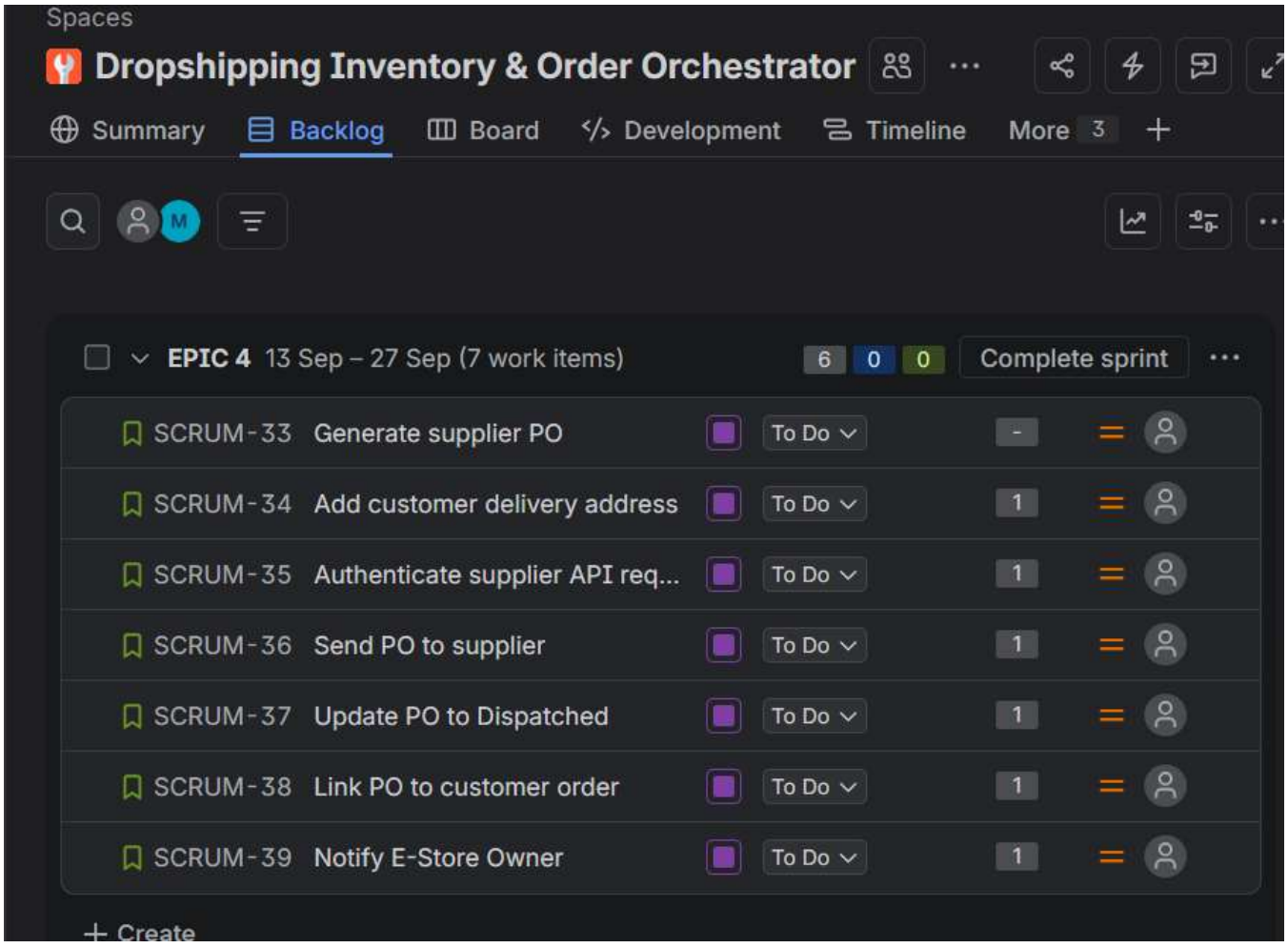
Completed work items — View in All work navigator

Key	Summary	Work type	Epic	Status	Assignee	Story points
SCRUM-12	Validate webhook signature	Story	Rec...	Done		1
SCRUM-16	Identify supplier SKUs	Story	Rec...	Done		1

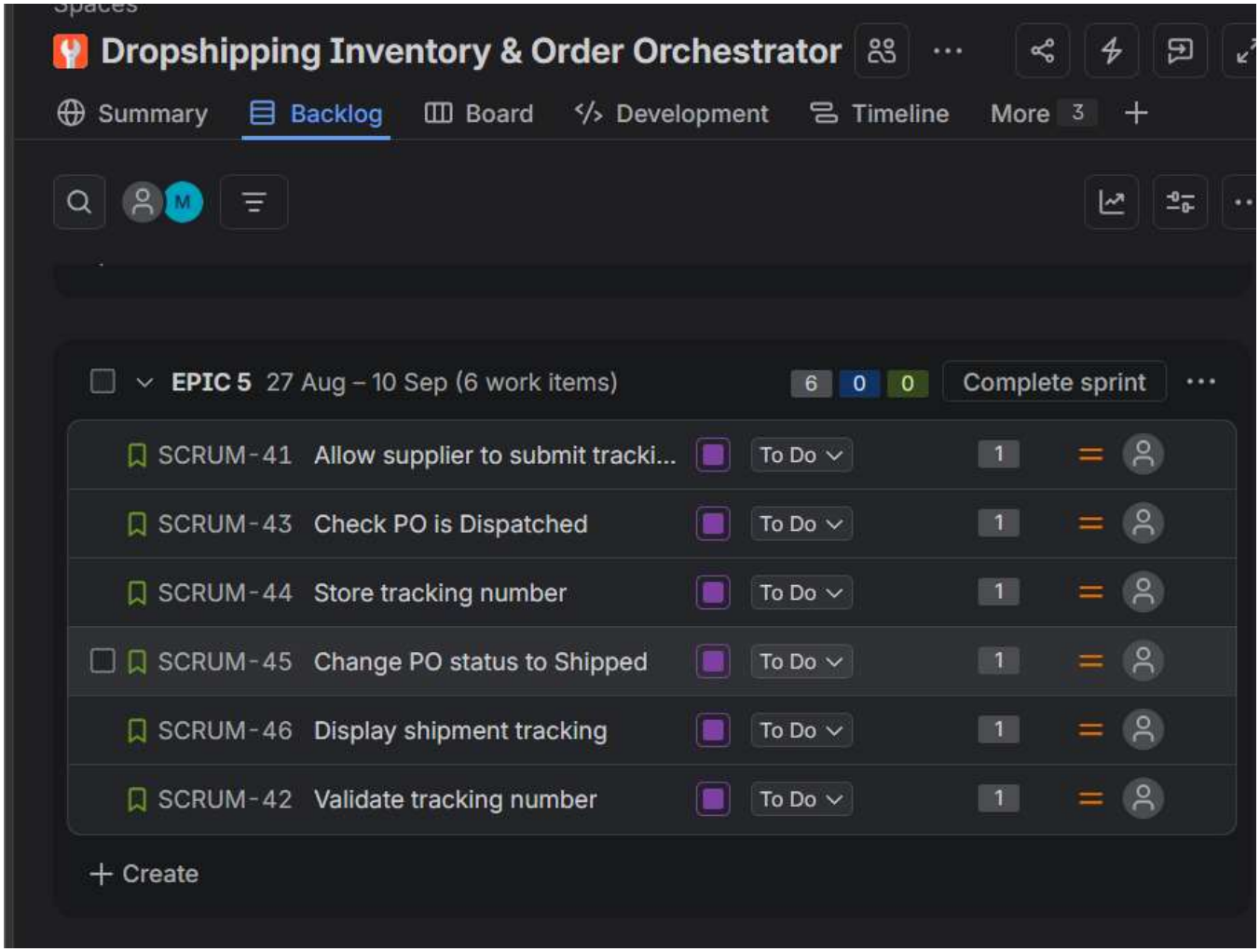
Story Point Evidence 1



Story Point Evidence 2



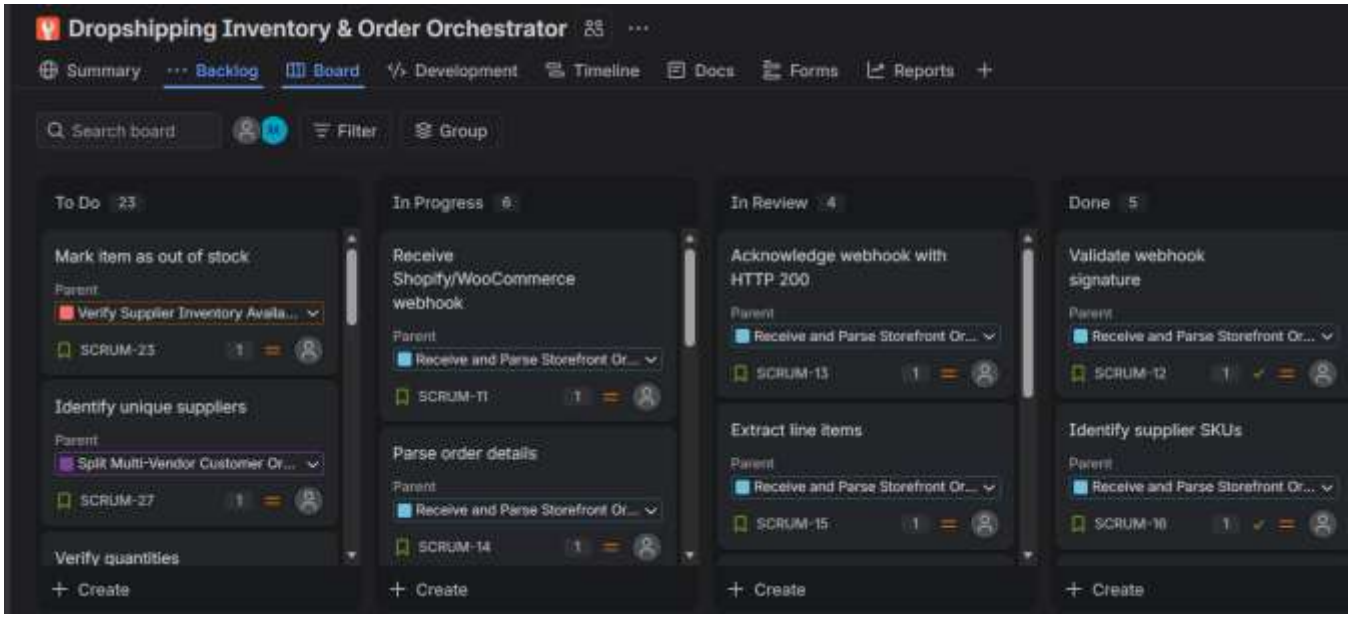
Story Point Evidence 4



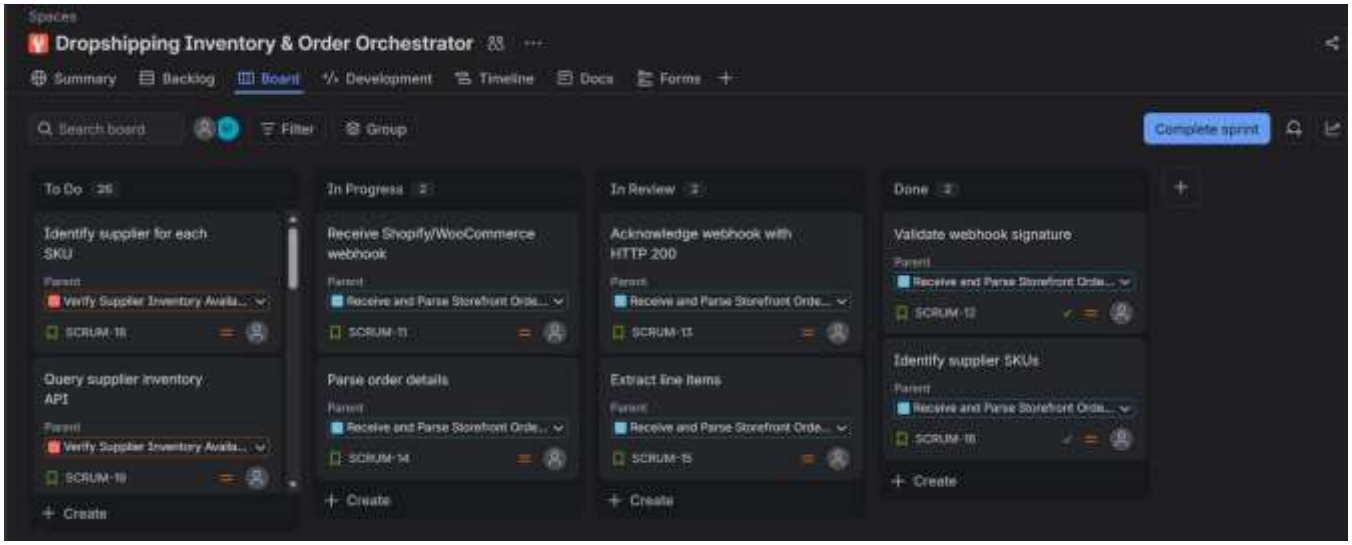
Story Point Evidence 5

4. Sprint Board – Active Sprint View

The Active Sprint board was used to simulate the Scrum workflow. User Stories were moved through To Do, In Progress, In Review, and Done, allowing the current sprint status to be observed.



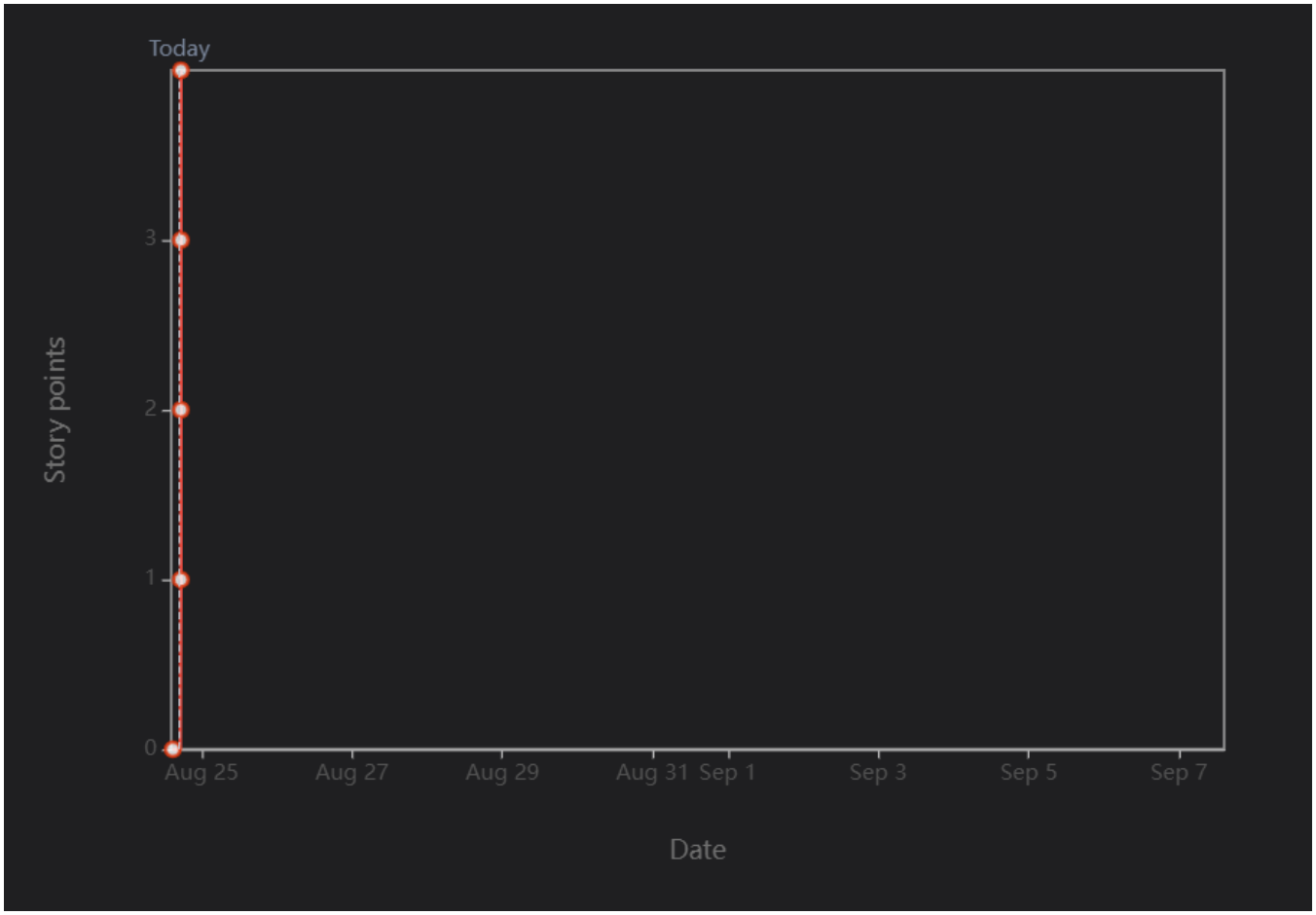
Active Sprint Board Evidence 1



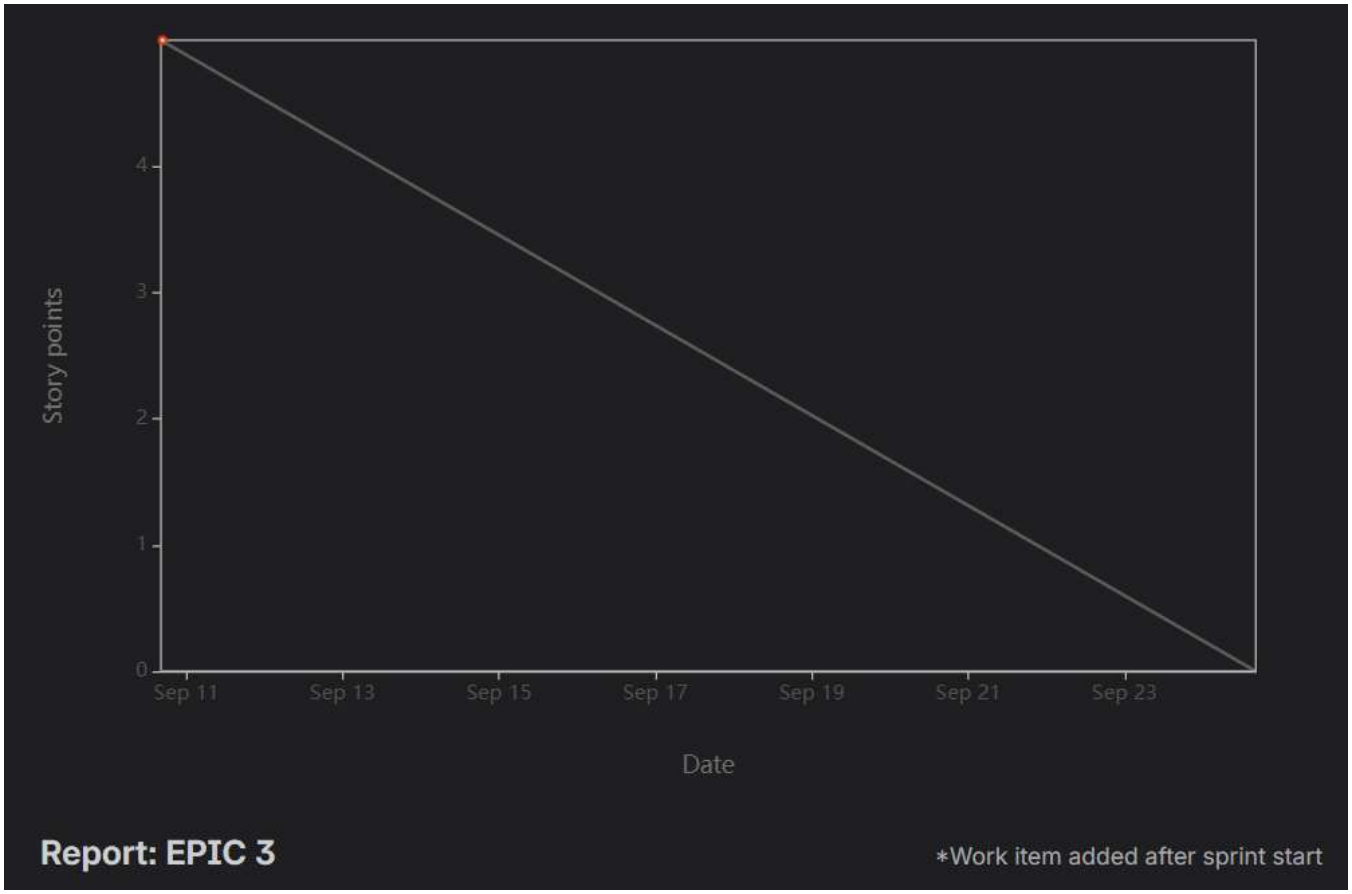
Active Sprint Board Evidence 2

5. Burndown Charts

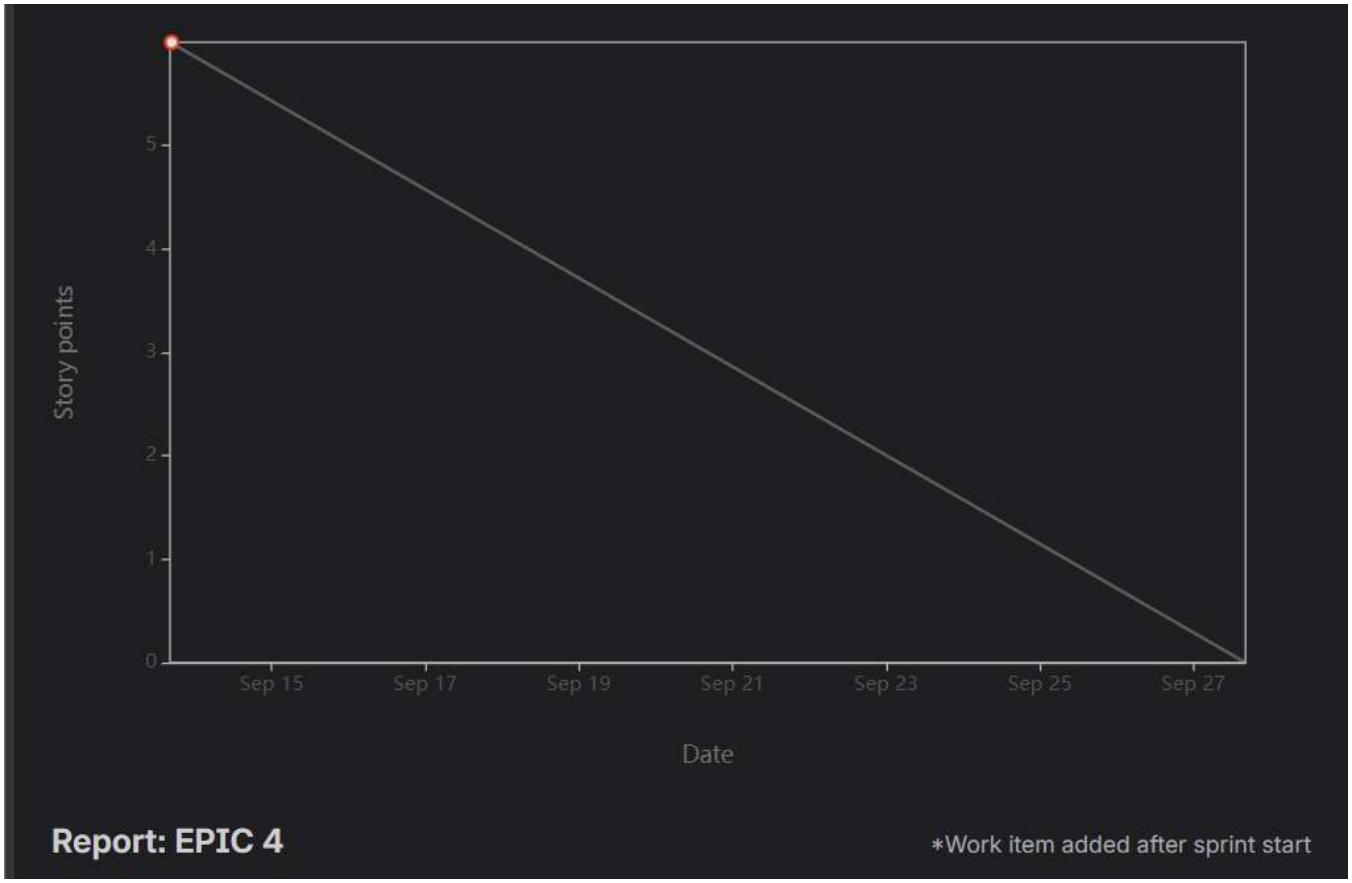
Jira Burndown Charts compare the remaining story-point work with the ideal burn-down guideline. The charts below document the configured sprint reports for the project's Epic-based sprints.



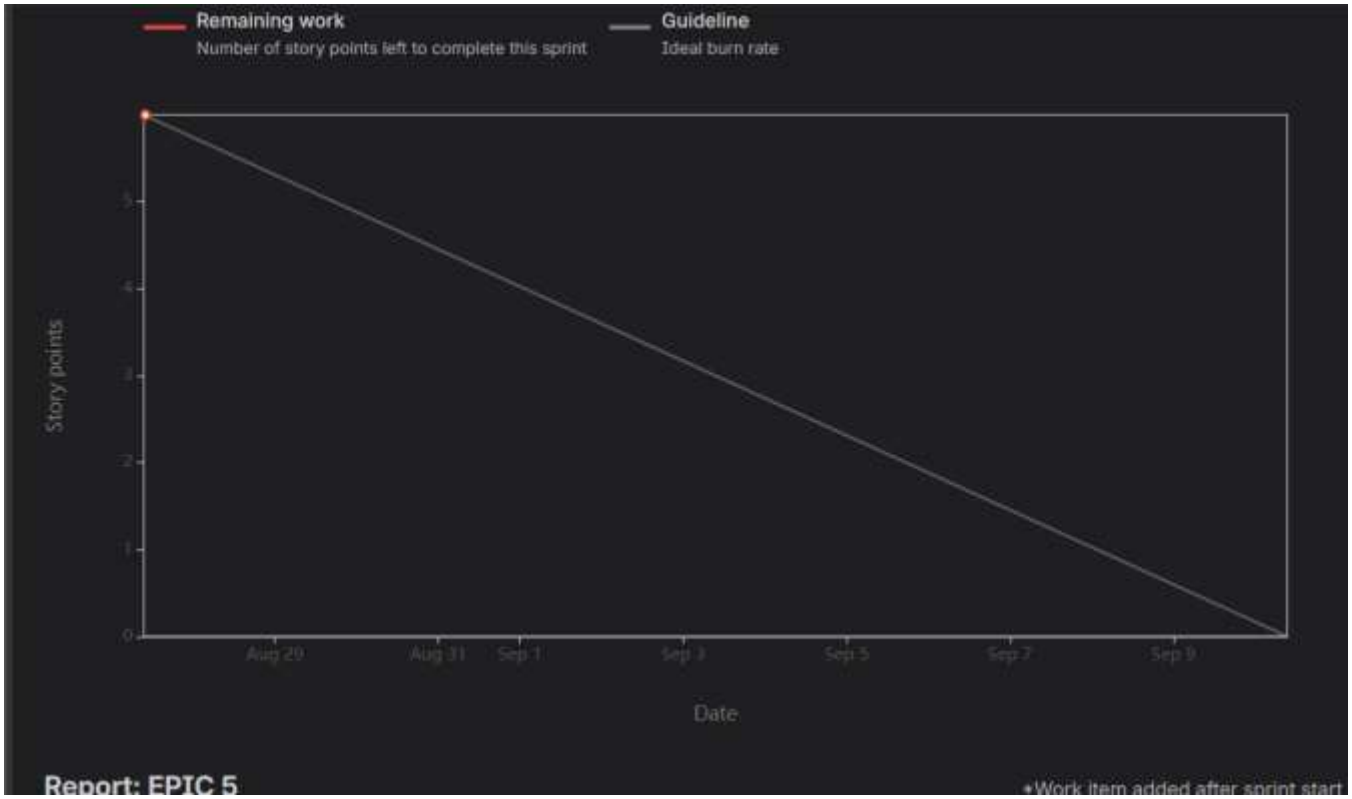
Burndown Chart Evidence 1



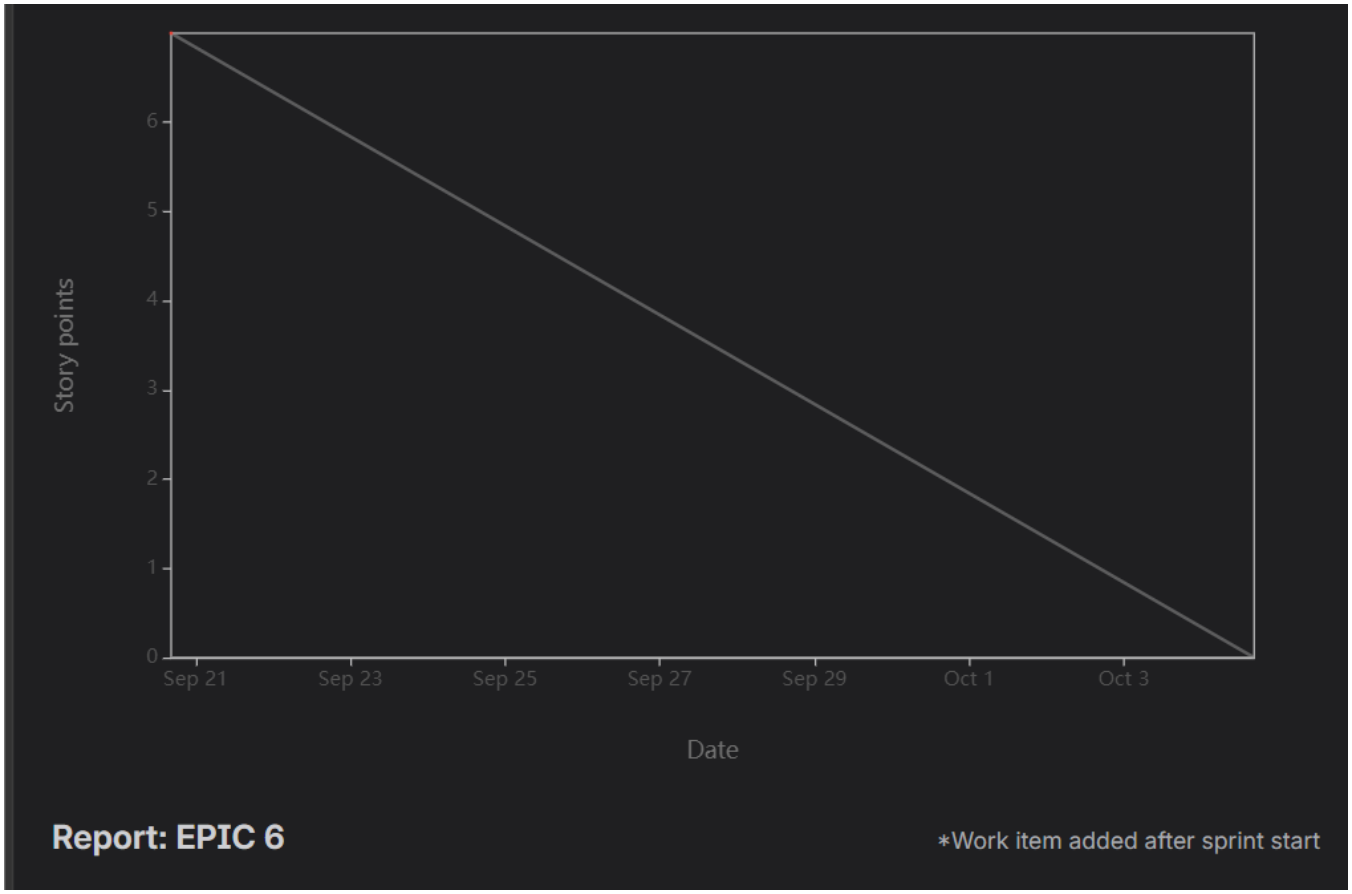
Burndown Chart Evidence 2



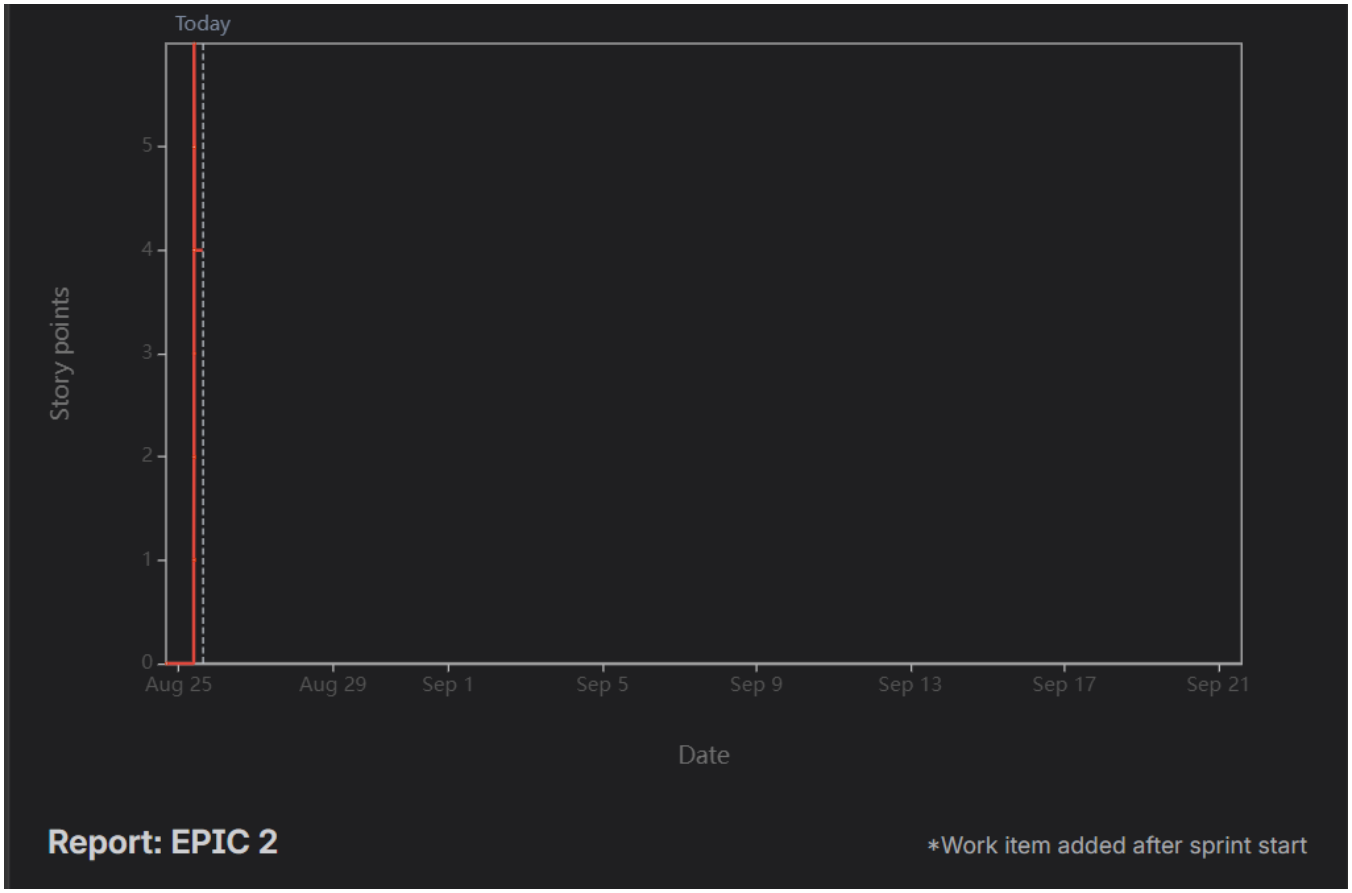
Burndown Chart Evidence 3



Burndown Chart Evidence 4



Burndown Chart Evidence 5



Burndown Chart Evidence 6

6. Reflection

The reflection questions below follow the structure of the provided SE Lab 2 reference document, with the responses adapted to the Dropshipping Inventory & Order Orchestrator project.

1. Did your estimations reflect the actual effort?

The story point estimations provided a reasonable representation of the effort involved in the different User Stories. The tasks were divided into smaller and focused activities, and each task was assigned 1 story point. This made the estimation process consistent across the sprints and helped in tracking the amount of work planned and completed. Since the project was carried out as a simulated sprint, the estimates mainly served as a means of comparing the relative workload of the different tasks.

2. Was your backlog well-prioritized?

Yes, the backlog was organized according to the functional flow of the Dropshipping Inventory & Order Orchestrator. The work was divided into logical Epics, beginning with receiving and parsing storefront orders, followed by supplier inventory verification, multi-vendor order processing, supplier purchase-order dispatch, shipment tracking, and order and fulfillment status. This organization provided a clear structure for managing the User Stories and helped maintain a logical sequence for the development of the system.

3. How did your simulated sprint align with your plan?

The simulated sprints followed the planned Scrum workflow in Jira. The User Stories were organized under their respective Epics and progressed through the workflow states To Do, In Progress, In Review, and Done. Separate sprints were used for the different functional areas of the project, which made it possible to track their progress independently. The Active Sprint board provided a clear view of the current status of the User Stories and helped in monitoring the overall progress of the project.

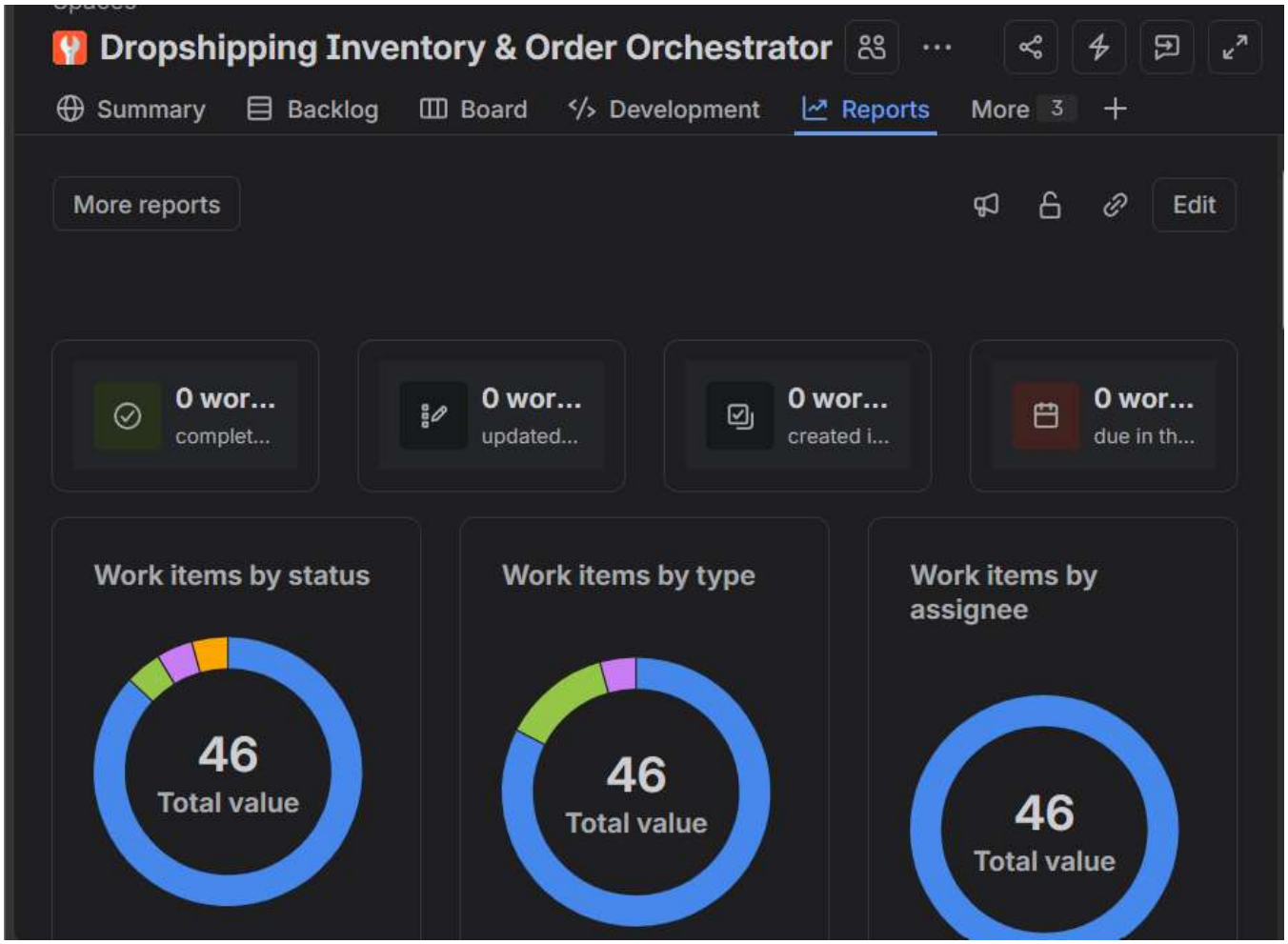
4. What insights did the burndown chart give about your team's capacity?

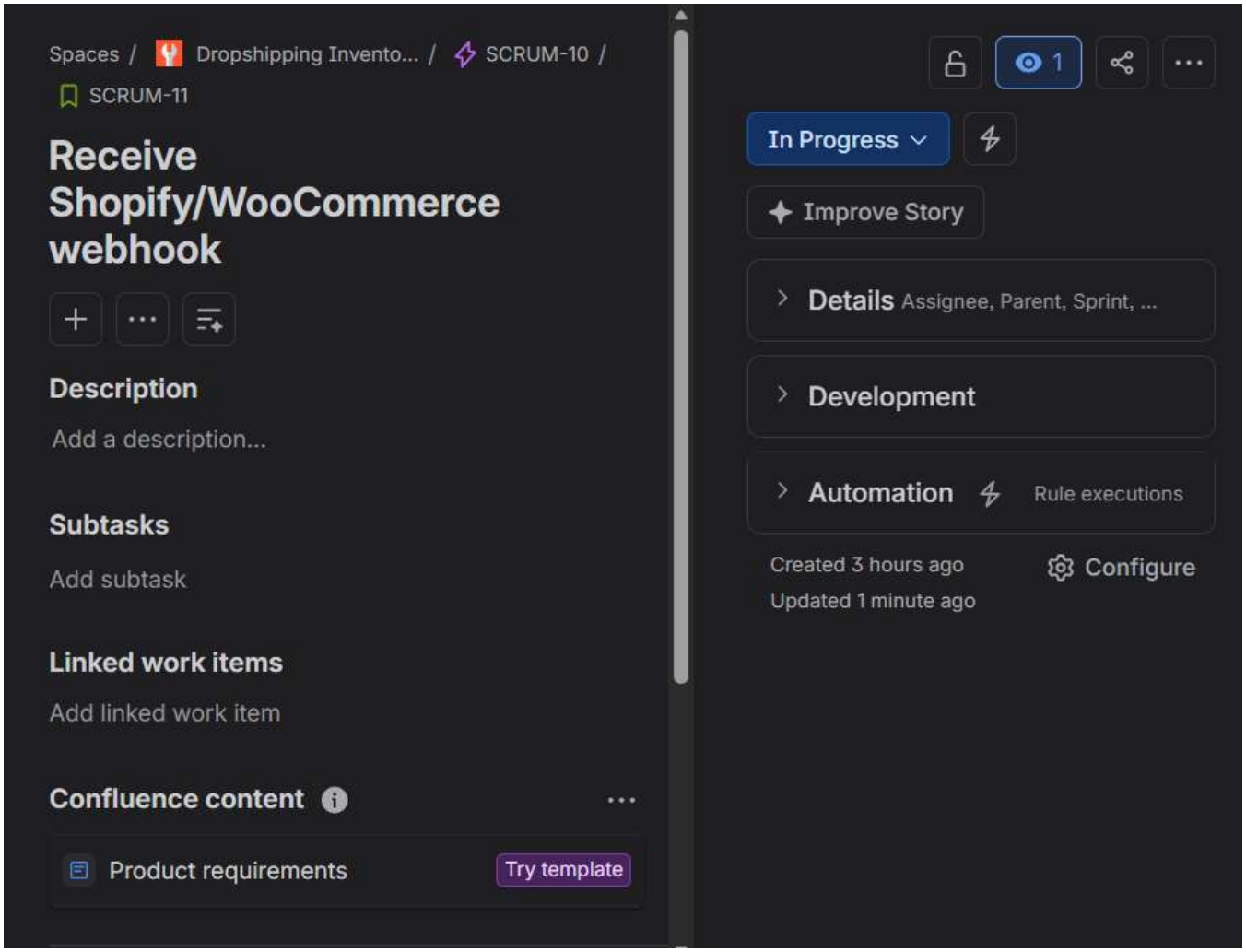
The burndown charts provided a clear representation of the remaining story-point work in each sprint in comparison with the ideal burn-down guideline. As User Stories were completed, the remaining workload was reduced. Since the project was a simulated Scrum exercise, the charts were primarily useful for understanding how the planned workload was distributed across the sprints and how task completion affected the remaining work. They also demonstrated the relationship between story-point estimation, sprint progress, and Jira's reporting features.

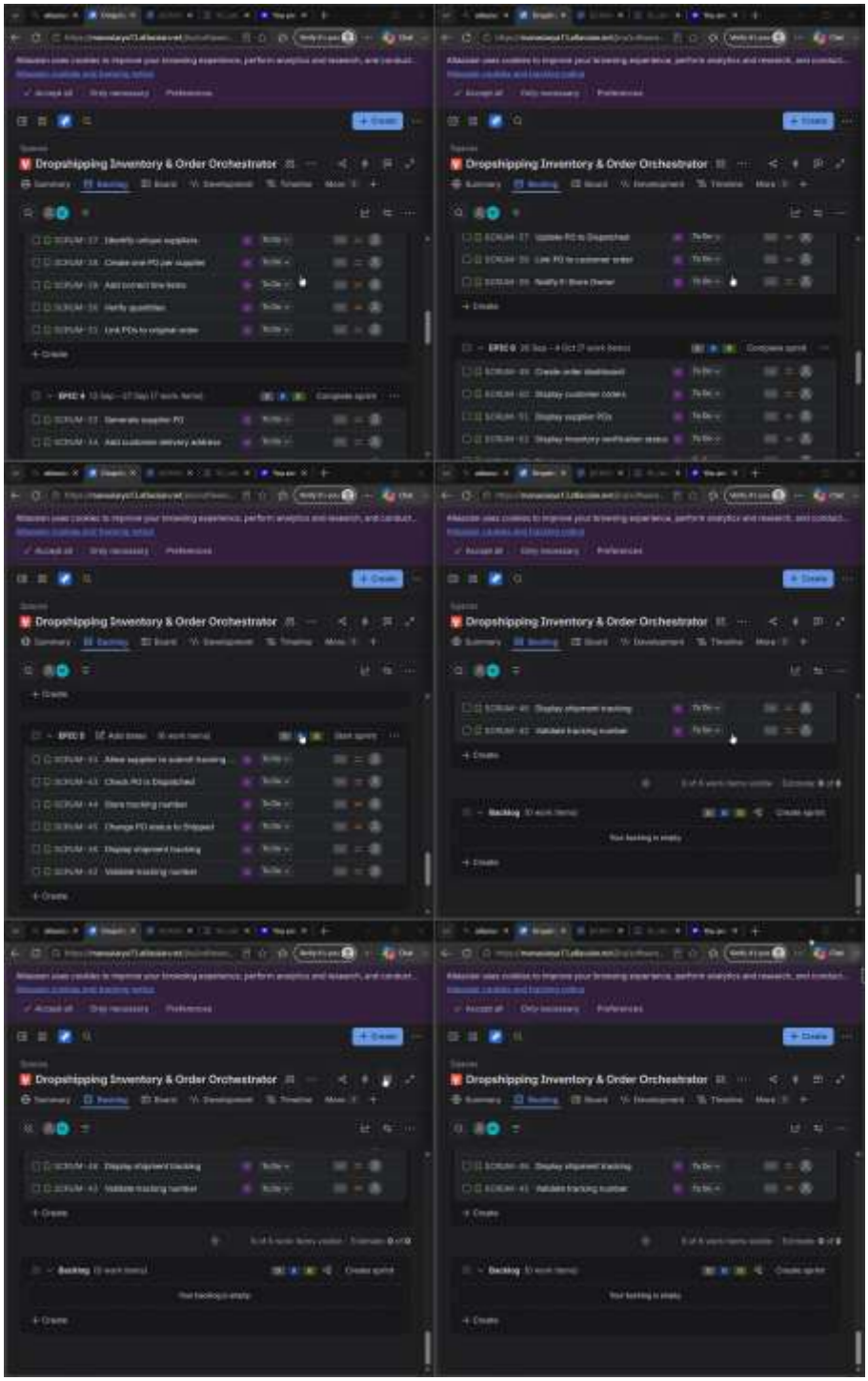
Appendix A – Additional Jira Screenshots

The following appendix preserves the additional Jira screenshots uploaded during the lab workflow, including configuration, work-item detail, navigation, and verification screens. They are included as supporting evidence in addition to the screenshots used for the main deliverables.

Additional Screenshot 1: ab8fda94-7ed4-46b0-a32a-fb4a7ed37156.png




Spaces / Dropshipping Invento... / SCRUM-10 /
SCRUM-11
Receive Shopify/WooCommerce webhook
Description
Add a description...
Subtasks
Add subtask
Linked work items
Add linked work item
Confluence content
Product requirements
Try template
1
In Progress
Improve Story
Details Assignee, Parent, Sprint, ...
Development
Automation
Rule executions
Created 3 hours ago
Updated 1 minute ago
Configure



Additional Screenshot 7: contact.png

Appendix B – Reference Used

The reflection-question structure was adapted from the provided reference document "SE_Lab 2 – Agile Backlog Creation & Sprint Simulation in Jira: Webhook Ingestion & Retry Mechanism Hub." The reference document contains four reflection questions covering estimations, backlog prioritization, sprint alignment, and burndown/capacity. The responses in this report were rewritten for the Dropshipping Inventory & Order Orchestrator project.